LN9 Completeness and Hedging

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Outlines

Textbook: Arbitrage Theory in Continuous Time: Chapter 8

1. Introduction
2. Completeness in the Black-Scholes Model
3. Completeness-Absence of Arbitrage

Introduction

Consider a price vector process $S = (S^1, \cdots, S^N)$ governed by an objective probability measure P. We now want to price a contingent T-claim $\mathcal{X}$.

Definition 1

We say that a T-claim $\mathcal{X}$ can be replicated, alternatively that it is reachable or hedgeable, if there exists a self-financing portfolio h such that

$$V^h(T) = \mathcal{X}, \ P - a.s.$$

In this case we say that h is a hedge against $\mathcal{X}$. Alternatively, h is called a replicating or hedging portfolio. If every contingent claim is reachable we say that the market is complete.

Introduction

Proposition 1

Suppose that the claim $\mathcal{X}$ can be hedged using the portfolio h. Then the only price process $\Pi(t;\mathcal{X})$ which is consistent with no arbitrage is given by $\Pi(t;\mathcal{X}) = V^h(t)$. Furthermore, if $\mathcal{X}$ can be hedged by g as well as by h then $V^g(t) = V^h(t)$ holds for all t with probability 1.

Completeness in the Black-Scholes Model

We will now investigate completeness for the generalized Black–Scholes model given by

$$\begin{aligned} dB(t) &= rB(t)dt, \\ dS(t) &= S(t)\alpha(t, S(t))dt + S(t)\sigma(t, S(t))d\bar{W}(t), \end{aligned} \tag{1}$$

Theorem 1

The Black-Sholes model is complete.

Completeness in the Black-Scholes Mode

Lemma 1

Suppose that there exists an adapted process V and an adapted process $u = [u^0, u^*]$ with $u^0(t) + u^*(t) = 1$, such that

$$\begin{aligned} dV(t) &= V(t)\left\{u^0(t)r + u^*(t)\alpha(t, S(t))\right\}dt + V(t)u^*(t)\sigma(t, S(t))d\bar{W}(t), \\ V(T) &= \Phi(S(T)). \end{aligned} \tag{2}$$

Then the claim $\mathcal{X} = \Phi(S(T))$ can be replicated using u as the relative portfolio. The corresponding value process is given by

$$h^0(t) = \frac{u^0(t)V(t)}{B(t)},$$

$$h^*(t) = \frac{u^*(t)V(t)}{S(t)}$$

Completeness in the Black-Scholes Mode

Theorem 2

Consider a contingent claim of the form $X = \Phi(S(T))$. Define F as the solution to the boundary value problem

$$\begin{aligned} F_t + rsF_s + \frac{1}{2}s^2\sigma^2 F_{ss} - rF = 0, \\ F(T,s) = \Phi(s). \end{aligned} \tag{3}$$

Then X can be replicated by the relative portfolio

$$u^0(t) = \frac{F(t,S(t)) - S(t)F_s(t,S(t))}{F(t,S(t))}$$

$$u^*(t) = \frac{S(t)F_s(t,S(t))}{F(t,S(t))}.$$

Theorem 2 Continued

Theorem 2 Continued

The corresponding absolute portfolio is given by

$$h^0(t) = \frac{F(t, S(t)) - S(t)F_s(t, S(t))}{B(t)}$$

$$h^*(t) = F_s(t, S(t)),$$

and the value process $V^h(t)$ is given by

$$V^h(t) = F(t, S(t)).$$

Completeness-Absence of Arbitrage

Theorem 3 (Meta-theorem)

Let M denote the number of underlying traded assets in the model excluding the risk free asset, and let R denote the number of random sources. Generically we then have the following relations:

1. *The model is arbitrage free if and only if $M \leq R$.*
2. *The model is complete if and only if $M \geq R$.*
3. *The model is complete and arbitrage free if and only if $M = R$.*